

Safe Drone Navigation Under Safety and Energy Constraints

1 Problem Statement

Consider a drone operating in a two-dimensional environment. The drone must travel from a given starting position to a target location while minimizing navigation error and control effort. During the mission, the drone must satisfy several safety requirements:

1. The drone must avoid entering a hazardous region.
2. The drone must not exceed a maximum allowable speed.
3. The total energy consumed throughout the mission must remain below a specified limit.
4. The control inputs must remain within physical actuator bounds.

The objective is to determine the optimal sequence of control actions that guides the drone safely to the destination while minimizing the overall mission cost.

2 Decision Variables

For a planning horizon of length T , the decision variables are the control inputs

$$a_{x,t}, \quad a_{y,t}$$

for

$$t = 0, 1, \dots, T - 1$$

where:

- $a_{x,t}$: acceleration command along the x-axis at time step t
- $a_{y,t}$: acceleration command along the y-axis at time step t

The control vector is defined as

$$u_t = \begin{bmatrix} a_{x,t} \\ a_{y,t} \end{bmatrix}$$

3 State Variables

The state of the drone at time step t is

$$s_t = \begin{bmatrix} x_t \\ y_t \\ v_{x,t} \\ v_{y,t} \end{bmatrix}$$

4 System Dynamics

$$x_{t+1} = x_t + v_{x,t}\Delta t$$

$$y_{t+1} = y_t + v_{y,t}\Delta t$$

$$v_{x,t+1} = v_{x,t} + a_{x,t}\Delta t$$

$$v_{y,t+1} = v_{y,t} + a_{y,t}\Delta t$$

5 Objective Function

$$J = \sum_{t=0}^{T-1} [(x_t - x_g)^2 + (y_t - y_g)^2 + \alpha(a_{x,t}^2 + a_{y,t}^2)]$$
$$\min J$$

6 Constraints

Energy constraint:

$$\sum_{t=0}^{T-1} (a_{x,t}^2 + a_{y,t}^2) \leq E_{\max}$$

Speed constraint:

$$v_{x,t}^2 + v_{y,t}^2 \leq V_{\max}^2$$

Hazard avoidance:

$$(x_t - x_h)^2 + (y_t - y_h)^2 \geq R_h^2$$

Control bounds:

$$a_{\min} \leq a_{x,t}, a_{y,t} \leq a_{\max}$$

7 Complete NLP Formulation

Minimize J subject to the dynamics and all constraints above.

8 Solution Using Lagrangian Optimization

Let λ_E be the multiplier associated with the energy constraint.

$$\mathcal{L} = J + \lambda_E \left(\sum_{t=0}^{T-1} (a_{x,t}^2 + a_{y,t}^2) - E_{\max} \right)$$

The solution procedure is:

1. Construct the Lagrangian.
2. Compute first-order derivatives.
3. Apply KKT conditions.
4. Solve the resulting nonlinear system.
5. Obtain optimal controls and multipliers.

9 Solution Using Sequential Quadratic Programming (SQP)

At iteration k :

1. Linearize all constraints.
2. Approximate the objective function using a quadratic model.
3. Form a QP subproblem.
4. Solve the QP.
5. Update the variables.
6. Repeat until convergence.

$$\min_d \quad \nabla J(x_k)^T d + \frac{1}{2} d^T H_k d$$

subject to

$$g(x_k) + \nabla g(x_k)^T d \leq 0$$

10 Expected Comparison

- Objective value
- Constraint satisfaction
- Runtime
- Number of iterations
- Convergence behavior